

Chapter 36: 17, 22, 27, 29, 37, 39

Basic score 4` + problem 29 2.5'+problem 37 2'+ problem 39 1.5'

17. (a) The first minimum of the diffraction pattern is at 5.00° ; so

$$a = \frac{\lambda}{\sin \theta} = \frac{0.620 \mu\text{m}}{\sin 5.00^\circ} = 7.11 \mu\text{m}$$

(b) Since the fourth bright fringe is missing, $d = 4a = 4(7.11 \mu\text{m}) = 28.44 \mu\text{m}$.

(c) For the $m = 1$ bright fringe,

$$\alpha = \frac{\pi a \sin \theta}{\lambda} = \frac{\pi (7.11 \mu\text{m}) \sin 1.25^\circ}{0.620 \mu\text{m}} = 0.786 \text{ rad}.$$

Consequently, the intensity of the $m = 1$ fringe is

$$I = I_m \left(\frac{\sin \alpha}{\alpha} \right)^2 = (7.0 \text{ mW/cm}^2) \left(\frac{\sin 0.786 \text{ rad}}{0.786} \right)^2 = 5.7 \text{ mW/cm}^2.$$

which agrees with Fig. 36-45. Similarly for $m = 2$, the intensity is $I = 2.9 \text{ mW/cm}^2$, also in agreement with Fig. 36-45.

22. (a) The slope of the plotted line is 12, and we see from Eq. 36-6 that this slope should correspond to

$$\frac{\pi a}{\lambda} = 24 \Rightarrow a = \frac{24\lambda}{\pi} = \frac{24(610 \text{ nm})}{\pi} = 4660 \text{ nm} \approx 4.66 \mu\text{m}$$

(b) Consider Eq. 36-3 with “continuously variable” m (of course, m should be an integer for diffraction minima, but for the moment we will solve for it as if it could be any real number):

$$m_{\text{max}} = \frac{a}{\lambda} (\sin \theta)_{\text{max}} = \frac{24\lambda}{\pi\lambda} = \frac{24}{\pi} \approx 7.6$$

which suggests that, on each side of the central maximum, there are seven minima; considering both sides then implies there are fourteen minima in the pattern.

(c) Setting $m = 1$, $a \sin \theta = \lambda$ yields $\theta = 7.5^\circ$.

(d) Setting $m = 7$, $a \sin \theta = 7\lambda$ yields $\theta = 66.4^\circ$.

27. We will make use of arctangents and sines in our solution, even though they can be “shortcut” somewhat since the angles are [almost] small enough to justify the use of the small angle approximation.

(a) Given $y/D = (0.350 \text{ m})/(4.00 \text{ m})$, then

$$\theta = \tan^{-1}\left(\frac{y}{D}\right) = \tan^{-1}\left(\frac{0.350 \text{ m}}{4.00 \text{ m}}\right) = 4.97^\circ = 0.087 \text{ rad}.$$

Equation 36-20 then gives

$$\beta = \frac{\pi d \sin \theta}{\lambda} = \frac{\pi (24.0 \mu\text{m}) \sin 4.97^\circ}{0.600 \mu\text{m}} = 10.9 \text{ rad}.$$

Thus, use of Eq. 36-21 (with $a = 12 \mu\text{m}$ and $\lambda = 0.60 \mu\text{m}$) leads to

$$\alpha = \frac{\pi a \sin \theta}{\lambda} = \frac{\pi (12.0 \mu\text{m}) \sin 4.97^\circ}{0.600 \mu\text{m}} = 5.5 \text{ rad}.$$

Thus,

$$\frac{I}{I_m} = \left(\frac{\sin \alpha}{\alpha}\right)^2 (\cos \beta)^2 = \left(\frac{\sin 5.5 \text{ rad}}{5.5}\right)^2 (\cos 10.9 \text{ rad})^2 = 1.50 \times 10^{-4}.$$

(b) Consider Eq. 36-25 with “continuously variable” m (of course, m should be an integer for interference maxima, but for the moment we will solve for it as if it could be any real number):

$$m = \frac{d \sin \theta}{\lambda} = \frac{(24 \mu\text{m}) \sin(4.97^\circ)}{0.6 \mu\text{m}} = 3.47$$

which suggests that the angle takes us to a point between the third side maximum (which would have $m = 3$) and the fourth minimum (which corresponds to $m = 3.5$), it almost locate at fourth minimum .

(c) Similarly, consider Eq. 36-3 with “continuously variable” m (of course, m should be an integer for diffraction minima, but for the moment we will solve for it as if it could be any real number):

$$m = \frac{a \sin \theta}{\lambda} = \frac{(12 \mu\text{m}) \sin(4.97^\circ)}{0.6 \mu\text{m}} = 1.74$$

which suggests that the angle takes us to a point between the first diffraction minimum ($m = 1$) and the second one ($m = 2$). The maxima (in the smaller peaks of the diffraction pattern) are not exactly midway between the minima; their location would make use of mathematics not covered in the prerequisites of the usual sophomore-level physics course.

29. (a) The angular positions θ of the bright interference fringes are given by $d \sin \theta = m\lambda$, where d is the slit separation, λ is the wavelength, and m is an integer. The first diffraction minimum occurs at the angle θ_1 given by $a \sin \theta_1 = \lambda$, where a is the slit

width. The diffraction peak extends from $-\theta_1$ to $+\theta_1$, so we should count the number of values of m for which $-\theta_1 < \theta < +\theta_1$, or, equivalently, the number of values of m for which $-\sin \theta_1 < \sin \theta < +\sin \theta_1$. This means $-1/a < m/d < 1/a$ or $-d/a < m < +d/a$. Now

$$d/a = (0.180 \times 10^{-3} \text{ m}) / (30.0 \times 10^{-6} \text{ m}) = 6.00 \quad 0.5 \text{ point}$$

so the values of m are $m = -5, -4, -3, -2, -1, 0, +1, +2, +3, +4$, and $+5$. There are 11 fringes. 0.5 point

(b) The intensity at the screen is given by

$$I = I_m \left(\cos^2 \beta \right) \left(\frac{\sin \alpha}{\alpha} \right)^2 \quad 0.5 \text{ point}$$

where $\alpha = (\pi a / \lambda) \sin \theta$, $\beta = (\pi d / \lambda) \sin \theta$, and I_m is the intensity at the center of the pattern. For the third bright interference fringe, $d \sin \theta = 3\lambda$, so $\beta = 3\pi$ rad and $\cos^2 \beta = 1$. Similarly, $\alpha = 3\pi a / d = 3\pi / 6.00 = 0.500\pi$ rad 0.5 point and

$$\left(\frac{\sin \alpha}{\alpha} \right)^2 = \left(\frac{\sin 0.500\pi}{0.500\pi} \right)^2 = 0.405 \quad 0.5 \text{ point}$$

The intensity ratio is $I/I_m = 0.405$.

37. The condition for a minimum of intensity in a single-slit diffraction pattern is $a \sin \theta = m\lambda$, where a is the slit width, λ is the wavelength, and m is an integer. To find the angular position of the first minimum to one side of the central maximum, we set $m = 1$:

$$\theta = \sin^{-1} \left(\frac{\lambda}{a} \right) = \sin^{-1} \left(\frac{650 \times 10^{-9} \text{ m}}{1.00 \times 10^{-3} \text{ m}} \right) = 6.5 \times 10^{-4} \text{ rad} \quad 0.5 \text{ point}$$

If D is the distance from the slit to the screen, the distance on the screen from the center of the pattern to the minimum is

$$y_1 = D \tan \theta_1 = (3.00 \text{ m}) \tan (6.5 \times 10^{-4} \text{ rad}) = 1.95 \times 10^{-3} \text{ m} \quad 0.5 \text{ point}$$

To find the third minimum, we set $m = 3$:

$$\theta = \sin^{-1} \left(\frac{3\lambda}{a} \right) = \sin^{-1} \left(\frac{3 \times 650 \times 10^{-9} \text{ m}}{1.00 \times 10^{-3} \text{ m}} \right) = 1.95 \times 10^{-3} \text{ rad}$$

The distance from the center of the pattern to this second minimum is

$$y_3 = D \tan \theta_3 = (3.00 \text{ m}) \tan (1.95 \times 10^{-3} \text{ rad}) = 5.85 \times 10^{-3} \text{ m} \quad 0.5 \text{ point}$$

The separation of the two minima is

$$y = y_2 - y_1 = 5.85 \text{ mm} - 1.95 \text{ mm} = 3.9 \text{ mm.} \quad 0.5\text{point}$$

Alternatively, we can use the small-angle approximation ($\tan \theta \approx \sin \theta \approx \theta$) for this problem because the angle θ is small enough

$$\Delta y = D(\tan \theta_3 - \tan \theta_1) \quad 0.5\text{point}$$

$$= D(\sin \theta_3 - \sin \theta_1) = D\left(\frac{3\lambda}{a} - \frac{\lambda}{a}\right) \quad 1 \text{ point}$$

$$= (3.00)\left(\frac{2 \times 650 \times 10^{-9}}{1.00 \times 10^{-3}}\right) = 3.9 \text{ mm} \quad 0.5\text{point}$$

39. (a) Using the notation of Sample Problem — “Pointillistic paintings use the diffraction of your eye,”

$$L = \frac{D}{1.22\lambda/d} \quad 0.5\text{point}$$

$$= \frac{2(50 \times 10^{-6} \text{ m})(1.8 \times 10^{-3} \text{ m})}{1.22(650 \times 10^{-9} \text{ m})} \quad 0.3 \text{ point}$$

$$= 0.23 \text{ m} \quad 0.2 \text{ point}$$

(b) The wavelength of the blue light is shorter so $L_{\max} \propto \lambda^{-1}$ will be larger. **0.5 point**